



Determination of Grain Yield and Quality Traits of Advanced Lines and Varieties of Oat (*Avena sativa* L.) in Kahramanmaraş Conditions

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ABSTRACT

This study was carried out in Kahramanmaraş conditions in the 2019-20 and 2020-21 growing seasons in order to investigate the grain yield and some quality traits of some advanced oat genotypes in terms of suitability for registration. The research was set up with three replications in randomized completeblock design. In the experiment, 8 advanced lines and 8 varieties (Kırklar, Kahraman, Küçükyayla, Halkalı, Yeniçeri, Sebat, Elmas and Aslanbey) in total 16 oat genotypes were used. In investigated traits, 1000 grain weight (TGW), test weight (TW), sieve analysis of grains (>2.2, 2.2-2.0, 2.0-1.5 mm), oil ratio (OR) and protein ratio (PR) were evaluated. According to the variance analysis results, genotype, year and genotype x year interaction were found to have significant effects on all traits except the 2.0-1.5 mm sieve analysis trait. A positive and significant relationship was determined between TGW and HLW ($r=0.6473^{**}$) and between TGW and >2.2 mm ($r=0.5710$). According to the two-year average of traits, TV; 238.9-425.8 (364.3) kg da⁻¹, TGW; 21.0-32.4 (28.9) g TW; 46.6-58.2 (52.7) kg hL⁻¹, >2.2 mm; 11.8-73.4% (53.2), 2.2-2.0 mm; 15.6-34.4% (23.3), 2.0-1.5 mm; 8.7-52.0% (22.8), OR; 0.28-6.38% (3.2) and PR; 9.4-11.8% (11.0) were determined. As a result of the study, the highest grain yield were observed in Kahraman (425.8 kg da⁻¹), Elmas (416 kg da⁻¹), line 7 (414.7 kg da⁻¹) and line 8 (405 kg da⁻¹). Kahraman and Elmas in varieties and 2, 11, 12 and 13 among advanced lines have also good quality characteristics. Advanced lines 7, 8, and 11 have been deemed suitable for registration based on all evaluated traits.

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Introduction

Oat, which is among the cereal crops, is a plant that has gained worldwide attention because it is a healthy and nutritious food. Although most of it is used in animal feeding due to its high protein ration it is also used in human consumption in the form of oatmeal and oat flour (Zhang et al., 2021). The good adaption of oats to different climatic conditions after wheat and barley allows oats to be produced reliably in marginal areas with harsh conditions, and plays an important role in the development of new generation crops to overcome food security challenges (Peng et al., 2021). interest in oat production and consumption has therefore slowly started to increase again compared to previous years (Kotikova et al., 2023). Oats (*Avena sativa* L.) are known to be a functional food source rich in protein, fat, carbohydrates, vitamins, and minerals (Flander et al., 2007). It is important for human health due to its high fibre content, which reduces the risk of certain diseases and heart attack by lowering cholesterol levels (Sarı et al., 2012). In human nutrition, oats are desired to have

high levels of protein, β -glucan, soluble dietary fiber, 1000 grain weight, hectoliter weight and internal aggregation of grains are high and fat and husks are low. For animal feeding, it is desired to have a 1000 grain weight, a hectoliter weight, a high protein and fat ratio, and a low β -glucan ratio (Peterson et al., 2005). While the protein content in hulled oats is 10-18%, it varies between 12-24% in hullless oats. Oats contain higher amounts of oil (3-12%) and unsaturated fatty acids than wheat and barley grains (Saastamoinen et al., 1992). Oats both hulls or without hulls can be stored healthily under optimal conditions (20°C temperature and 12-14% humidity) for a year as long as the grain integrity is not compromised. Oats and oat products have a short shelf life due to the high amount of fat, a significant portion of the fatty acids being unsaturated fatty acids (oleic acid and linoleic acid) and having 10-15 times higher lipase activity than wheat (Matlashewski et al. 1982). The number and size of grains in the panicle vary according to the number of spikelets in the panicle and the

number of grains in the panicle. The number of grains in a cluster varies between 40 and 200 depending on the variety, climatic conditions and planting density (Yürür, 1994). Oat grains are hulled like barley, and the grains are not homogeneous. Hull content in oats is an important quality criterion affecting grain yield and nutritional value. The husk ratio in oats varies between 17.0-39.5% (28.1) depending on the varieties and growing conditions (Kahraman et al., 2017b). Although the oat plant has an important value in terms of animal and human health and is well adapted to the climatic conditions of Türkiye's geography, its cultivation is less than wheat and barley. In Türkiye, oats were planted in 137655 ha of grain area, 360719 ha of green grass area and a total area of 498375 ha. From these planted areas, 365000 tons of grain and 4649051 tons of green grass oat production was realized in 2022 (TURKSTAT, 2023).

Therefore, in this research, it was aimed to determine the suitable genotypes for Kahramanmaraş region and to increase oat cultivation in the region by examining the grain yield and quality characteristics of some oat genotypes in Kahramanmaraş conditions.

Materials and Methods

The study was conducted during 2 years in the 2019-20 and 2020-21 growing seasons in Kahramanmaraş Sütçü İmam University, Faculty of Agriculture, Field Crops Research and Application field. In the study, there were used 8 check oat varieties and 8 advanced oat lines from the Thrace Agricultural Research Institute Directorate

were used. The material list of oat genotypes used in the experiment is given in Table 1.

The experiment was conducted according to the randomized complete block design with three replications. The trial materials were sown on 8 November 2019 in the first year and on 18 November 2020 in the second year. Every plot had 6 m x 1 m = 6 m² areas and 500 seeds per m². seeds were sown using precision experiment seeder. 7 kg N da⁻¹ and 7 kg P da⁻¹ per decare (18.46.0 DAP) chemical nutrient were given to soil at the time of sowing, and 7 kg N da⁻¹ (46% N Urea) nutrient was applied as spring fertilizier at the stalking period. For weed control, herbicide (2,4-D amin) was applied on 8 March in 2020 and on 6 April in 2021. Climate data from the trial years in Kahramanmaraş province are given at Table 2. According to Table 2, total rainfall in the first year growing season (652.3 mm) was higher than in the second year (512.6 mm). In the region, the amount and distribution of rainfall in March and April are more effective on grain yield for oat. Plants are fully mature at the end of May and beginning of June. Due to the high amount of rainfall in the first year and the high temperature in the second year, the grain yields of oat genotypes were higher than in the second year. Due to the high temperature in the second year, the plants matured earlier and the harvest was done 5 days earlier.

According to the soil analysis results of the research site where the study was carried out, it was seen that the soil had a clayey loam structure (52.8%), the pH value (6.92) was close to neutral, the amount of lime (0.20%) was very low, the amount of organic matter (2.61%) was moderate, and it had sufficient potassium (218.90 mg kg⁻¹) and phosphorus content (15.14 mg kg⁻¹) (Anonymous, 2021b).

Table 1. Material list of oat genotypes used in the experiments

Reference	Variety / Pedigree	Reference	Variety / Pedigree
1	Küçükyayla	9	LA09092SBS-0BD-0BD-0T-14T-0T
2	Kahraman	10	LA09097SBS-0BD-0BD-0T-6T-0T
3	Kırklar	11	TEY 008-0T-0T-8T-0T
4	Halkalı	12	TEY 014-0T-0T-2T-0T
5	Yeniçeri	13	Elmas
6	Sebat	14	TEY 027-0T-0T-3T-0T
7	LA09088SBS-0BD-0BD-0T-2T-0T	15	TEY 030-0T-0T-5T-0T
8	LA09088SBS-0BD-0BD-0T-5T-0T	16	Aslanbey

Table 2. The climate values for Kahramanmaraş province for the 2019-20 and 2020-21 crop seasons

Months	Years									
	Average Relative Humidity (%)		Total Precipitation (mm)		Average Temperature (°C)		Highest Temperature (°C)		Lowest Temperature (°C)	
	I.Y	II.Y	I.Y	II.Y	I.Y	II.Y	I.Y	II.Y	I.Y	II.Y
November	56.2	65.9	39.1	55.4	13.5	12.4	25.5	25.4	5.0	3.4
December	81.9	74.7	198.5	56.8	8.4	8.2	17.8	17.0	3.6	-0.9
January	69.3	70.2	88.0	211.6	6.3	7.0	14.2	17.4	-1.0	-2.3
February	68.3	59.8	72.7	29.6	6.1	9.3	18.2	18.7	-6.7	-2.3
March	67.3	61.2	173.4	129.2	12.5	10.4	23.8	21.4	1.0	2.1
April	58.2	57.5	61.8	16.8	15.9	16.6	27.9	31.0	4.3	4.6
May	47.2	43.3	18.5	13.2	21.6	23.5	38.1	36.0	9.8	12.4
June	46.9	49.0	0.3	0.0	21.7	25.5	28.0	40.4	14.7	14.5
Total	--	--	652.3	512.6	--	--	--	--	--	--
Average	61.91	60.20	--	--	13.25	14.11	24.19	25.91	3.84	4.22

I.Y- 2019-2020; II.Y-2020-2021

Table 3. Combined analysis of variance for grain yield (GY), thousand grain weight (TGW), test weight (TW) and >2.2 mm sieve of oat genotypes

Source of Variation	df	GY (kg da ⁻¹)		TGW (g)		TW ((kg hL ⁻¹)		.2 mm Sieve Top >2.2 mm	
		Mean Squares	P Value	Mean Squares	P Value	Mean Squares	P Value	Mean Squares	P Value
Year (Y)	1	3683045*	0.0414	111.80**	0.0073	566.09**	0.0001	345.04**	0.0091
Block [Year]	4	419563	0.0001	4.39	0.0002	0.34	0.1903	15.43	0.0113
Genotype(G)	15	597780**	0.0001	68.79**	0.0001	48.20**	0.0001	2424.56**	0.0001
Y × G	15	183310**	0.0010	14.38**	0.0001	4.26**	0.0001	208.68**	0.0001
Error	60	59683		0.66		0.21		4.33	
Total	95	217459.3		14.91		14.39		422.79	

*: p<0.05, **: p<0.01 significant

Table 4. Combined analysis of variance for 2.2-2.0 mm and 2.0-1.5 mm sieves, oil ratio (OR) and crude protein ratio (PR) of oat genotypes

Source of Variation	df	2.2-2.0 mm between the sieves		2.0-1.5 mm between the sieves		OR (%)		PR(%)	
		Mean Squares	P Value	Mean Squares	P Value	Mean Squares	P Value	Mean Squares	P Value
Year (Y)	1	213.31**	0.0001	4.99594	0.6115	10.41**	0.0032	61.76**	0.0001
Block [Year]	4	0.60	0.8477	16.5065	0.0001	0.26	0.0049	0.03	0.3563
Genotype(G)	15	248.83**	0.0001	1259.7**	0.0001	18.46**	0.0001	2.76**	0.0001
Y × G	15	58.04**	0.0001	120.752**	0.0001	3.04**	0.0001	1.63**	0.0001
Error	60	1.75		2.04		0.06		0.03	
Total	95	51.83		219.99		3.55		1.36	

*: p<0.05, **: p<0.01 significant

Harvesting was done using a special plot combine harvester when the plants reached physiological maturity on 12 June 2020 in the first year and on 07 June 2021 in the second year. The traits to be examined were made separately for each plot and grain yield (GY), thousand grain weight (TGW), test weight (TW), >2.2 mm sieve top, 2.2-2.0 mm, 2.0-1.5 mm, oil ratio (OR) and protein ratio (PR) traits were examined. Grain yields were calculated as kilograms per decare according to the plots. Sample grains from each plot were ground into flour and subjected to oil analysis. Protein analysis was also performed by grinding sample grains from each plot into flour. Protein content (nitrogen content * 5.83) was determined using the AOAC 992.23 method and a Perten IM:9500 assay device (Anonymous, 2000).

Grain yields were calculated as kilograms per decare according to the plots. The findings obtained as a result of the study were subjected to variance analysis with the help of the JMP statistical package program. According to the variance analysis results, the statistically significant factor means were compared with the LSD test (Kalaycı, 2005).

Results and Discussion

The study was carried out to determine the grain yield and quality characteristics of 16 different oat (*Avena sativa* L.) genotypes under Kahramanmaraş conditions. Combined ANOVA results are presented in Tables 3–4. Trait means are presented in Tables 5–7.

There were significant differences among years (p<0.05), genotypes (p<0.01) and genotype x year interaction (p<0.01) in grain yield trait of genotypes. In the study, the grain yield of the genotypes varied between 262.9-475.4 (396.9) kg da⁻¹ in the 1st year, 211.0-411.4

(331.6) kg da⁻¹ in the 2nd year, and 238.9-425.8 (364.3) kg da⁻¹ according to the two-year average. According to the two-year average, genotype number 2 had the highest grain yield with 425.8 kg da⁻¹ yield, while genotype number 6 had the lowest grain yield with 238.9 kg da⁻¹ yield (Table 5). According to the data obtained in the research, there was a decrease in the grain yield -in the second year compared to the first. As stated in the studies of Tamm (2003) and Peterson et al., (2005), the effect of climate factors as well as oat genotypes affects grain yield. As seen in Table 2, the higher average temperature value, lower relative humidity and lower total rainfall in the second year compared to the first year negatively affected grain yield. In previous studies, Doehlert et al. (2001) found that grain yield in ten oat genotypes varied between 314 kg da⁻¹ and 411 kg da⁻¹. Kahraman et al. (2017a) reported that oat genotypes created statistically significant differences in their study conducted in different locations for 2 years, and grain yield among genotypes varied between 372.1 kg da⁻¹ and 734.8 kg da⁻¹. This indicates that grain yield will vary depending on genotypes, environmental conditions and growing conditions. Halil and Uzun (2019) found that genotypes, years and year x genotype interaction on grain yield were statistically significant and that the grain yield was between 306.71-454.84 kg da⁻¹, thus supporting our findings. Georgieva and Zorovski (2019) found that grain yield in oat genotypes was between 332.55 kg da⁻¹ and 427.26 kg da⁻¹. Howarth et al. (2021) reported grain yields ranging from 4.84 t ha⁻¹ to 10.49 t ha⁻¹ in four winter oat cultivars at different locations over four years.

A significant difference (p<0.01) were determined among years, genotypes and genotype x year interaction in terms of 1000 grain weight (Table 3). According to the research findings, 1000 grain weight of the genotypes varied between

22.1-34.0 (29.9) g in the first year, 19.8-32.6 (27.8) g in the second year, and 21.0-32.4 (28.9) g according to the two year average. According to the two-year average, genotype 11 reached the highest 1000-grain weight with 32.4 g, and genotype 6 followed with 21.0 g (Table 5). The 1000 grain weight of the genotypes in the first year was higher than in the second year. In previous studies, Brunava et al. (2015) reported that 1000 grain weight varied between 36.98 g and 39.90 g in 19 oat varieties as a result of 3 year study. Kahraman et al. (2017a) emphasized that 1000 grain weight should be higher than 26 g for human nutrition in oat genotypes tested in different locations for 2 years and were found between 18.7 g and 45.0 g among the genotypes. Our findings have partially supported by their reports. Georgieva and Zorovski (2019) reported in their study that 1000 grain weight in oat genotypes varied between 26.58 g and 32.12 g. Halil and Uzun (2019) found that genotypes, years and year x genotype interactions were statistically significant on 1000-grain weight and that 1000-grain weight was between 24.25 g and 37.84 g, partially supporting our findings. Howarth et al. (2021) reported that 1000 grain weight of four winter oat cultivars in different locations over four years ranged from 35.52 g to 48.18 g. Kahraman et al. (2021) reported that 1000 grain weight varied between 21.1 g and 41.3 (33.5) g in 14 oat genotypes. Mut et al. (2021) reported that the thousand-grain weight of the genotypes ranged from 24.1 to 38.5 g under rainfed conditions and from 24.2 to 39.3 g under irrigated conditions. Güngör et al. (2023) observed that the average 1000 grain weight varied between 26.6 g and 40.0 g in oat genotypes at different locations during a two year study.

In terms of test weight, there significant differences were determined among years, genotypes and genotype x year interaction ($p < 0.01$) (Table 3). According to the two-year average, genotype 2 reached the highest test weight with 58.2 kg hL⁻¹, while genotype 6 reached the lowest test weight with 46.6 kg hL⁻¹ (Table 5). The test weight of the genotypes in the first year was approximately 10% higher than the second year. In previous studies, Kahraman et al. (2017a) supported our findings by stating that the test

weight was between 43.9 kg and 60.4 kg in oat genotypes tested in different locations in 2 years. Georgieva and Zorovski (2019) supported our findings by reporting that the test weight in oat genotypes varied between 46.38 kg and 53.23 kg in their study. Howarth et al. (2021) reported that test weight of four winter oat cultivars varied between 42.64 kg hL⁻¹ and 59.57 kg hL⁻¹ in different locations over four years, which is in agreement with our findings. Kahraman et al. (2021) reported that test weight varied between 47.0 kg hL⁻¹ and 59.8 kg hL⁻¹ in 14 oat genotypes. Hocaoglu et al. (2022) reported that test weight varied between 44.17 kg hL⁻¹ and 55.65 kg hL⁻¹ in oat genotypes, which supports our findings.

When Table 3 was examined, a statistically significant difference was determined between the 2.2 mm sieve top (>2.2) value of the genotypes and year, genotype and genotype x year interaction ($p < 0.01$). In the study, the 2.2 mm sieve top value of the genotypes varied between 17.2-72.1% (51.3) in the 1st year, 6.3-81.2% (55.1) in the 2nd year, and 11.8-74.2% (53.2) according to the two-year average. According to the two-year average, genotype no. 11 with 74% reached the highest 2.2 mm, and genotype no. 16 with 11.8% reached the lowest 2.2 mm sieve top (>2.2) value (Table 6). In the first year, the 2.2 mm sieve value of the genotypes was lower than the second year, unlike grain yield, 1000 grain weight and hectoliter weight. The number of grains in the spikelet of oats varies between 2-3 on average, and as the number of grains in the cluster increases, the grain size decreases. In previous studies, Brunova et al. (2015) reported that the 2.5 mm sieve top value varied between 58.17% and 61.68% and the 2.2-2.5 mm sieve value varied between 30.96% and 34.85% in 19 oat varieties as a result of a 3-year study. Sari et al. (2016) determined the 2.2 mm sieve top value (>2.2) between 26.0 mm and 83.5 mm in oat genotypes. Kahraman et al. (2017a) reported that the 2.2 mm sieve (>2.2) value in oat genotypes tested in different locations for 2 years was between 17.3% and 95.5%.

Table 5. Grain yield, 1000 grain and hectolitre weight values of oat genotypes for 2 years

G. No.	Grain yield (kg da ⁻¹)			1000 grain weight (g)			Hectolitre weight (kg hL ⁻¹)		
	2019-20	2020-21	2019-21	2019-20	2020-21	2019-21	2019-20	2020-21	2019-21
1	435.1±7.8 ^{ab}	335.0±27.4 ^c	385.0±25.8 ^{a-c}	32.6±0.4 ^{bc}	31.1±0.9 ^{a-c}	31.9±0.5 ^{ab}	56.9±0.3 ^{cd}	51.8±0.3 ^d	54.3±1.2 ^c
2	475.4±30.3 ^a	376.2±26.2 ^{a-c}	425.8±28.5 ^a	32.9±0.4 ^{a-c}	28.7±0.8 ^e	30.8±1.0 ^{cd}	60.2±0.2 ^a	56.2±0.3 ^a	58.2±0.9 ^a
3	379.1±32.2 ^{b-d}	411.4±8.2 ^a	395.3±26.5 ^{a-c}	31.4±0.1 ^{de}	29.3±0.7 ^{de}	30.3±0.6 ^{de}	54.8±0.2 ^g	49.6±0.1 ^{fg}	52.2±1.2 ^{ef}
4	383.2±41.1 ^{b-d}	360.4±10.5 ^{a-c}	371.8±19.6 ^{bc}	26.9±0.1 ^g	24.5±0.1 ^{gh}	25.7±0.5 ^h	53.9±0.2 ^h	49.8±0.3 ^{efg}	51.8±0.9 ^{fg}
5	405.7±29.3 ^{a-d}	341.2±28.4 ^c	373.5±23.3 ^{bc}	22.6±0.3 ⁱ	22.4±0.1 ⁱ	22.5±0.1 ⁱ	52.9±0.3 ⁱ	50.4±0.1 ^{ef}	51.7±0.6 ^g
6	262.9±15.2 ^e	215.0±21.9 ^{ef}	238.9±16.0 ^e	22.1±0.4 ⁱ	19.8±0.6 ^j	21.0±0.6 ^j	50.1±0.2 ^j	43.1±0.1 ^k	46.6±1.6 ⁱ
7	462.9±26.1 ^a	366.5±17.1 ^{a-c}	414.7±25.6 ^{ab}	32.2±0.5 ^{b-d}	29.7±0.8 ^{c-e}	31.0±0.7 ^{b-d}	57.1±0.2 ^c	53.2±0.4 ^b	55.2±0.9 ^b
8	466.4±20.8 ^a	343.7±20.1 ^c	405.0±30.3 ^{a-c}	25.7±0.5 ^h	32.6±0.7 ^{2a}	29.2±1.6 ^f	56.2±0.3 ^{de}	52.9±0.3 ^{bc}	54.6±0.7 ^c
9	413.6±38.7 ^{a-c}	211.0±31.1 ^f	312.3±50.4 ^d	28.5±0.4 ^f	26.7±0.1 ^f	27.6±0.4 ^g	53.3±0.3 ^{hi}	50.5±0.3 ^e	51.9±0.7 ^{fg}
10	331.1±42.2 ^{de}	270.2±7.4 ^{de}	300.7±23.5 ^d	32.2±0.3 ^{bcd}	23.4±0.1 ^{hi}	27.8±2.0 ^g	56.3±0.3 ^{de}	49.2±0.5 ^{gh}	52.7±1.6 ^e
11	430.4±9.4 ^{ab}	354.9±42.6 ^{bc}	392.7±25.8 ^{a-c}	33.3±0.1 ^{ab}	31.6±0.9 ^{ab}	32.4±0.5 ^a	58.0±0.1 ^b	53.0±0.4 ^{bc}	55.5±1.1 ^b
12	376.4±46.0 ^{b-d}	374.0±12.3 ^{a-c}	375.2±21.3 ^{bc}	33.1±0.3 ^{a-c}	30.7±0.6 ^{b-d}	31.9±0.6 ^{ab}	56.0±0.3 ^{ef}	52.2±0.1 ^{cd}	54.1±0.9 ^c
13	422.4±50.8 ^{ab}	409.6±17.0 ^{ab}	416.0±24.1 ^{ab}	32.1±0.7 ^{cd}	30.4±1.1 ^{b-d}	31.3±0.7 ^{b-d}	56.6±0.3 ^{c-e}	50.3±0.1 ^{ef}	53.5±1.4 ^d
14	403.3±15.3 ^{a-d}	323.2±14.2 ^{cd}	363.2±20.2 ^c	30.4±0.44 ^e	28.6±0.67 ^e	29.5±0.53 ^{ef}	55.3±0.33 ^{fg}	48.3±0.17 ⁱ	51.8±1.58 ^{fg}
15	336.9±36.3 ^{c-e}	238.1±21.7 ^{ef}	287.5±29.1 ^d	29.2±0.3 ^f	25.9±0.9 ^{fg}	27.5±0.8 ^g	49.1±0.1 ^k	46.0±0.3 ^j	47.6±0.7 ^h
16	365.7±23.3 ^{b-d}	375.5±27.2 ^{a-c}	370.6±16.2 ^{bc}	34.0±0.1 ^a	29.2±0.1 ^{de}	31.6±1.1 ^{a-c}	56.0±0.2 ^{ef}	48.4±0.2 ^{hi}	52.2±1.7 ^{ef}
M	396.9±9.9 ^a	331.6±10.4 ^b	364.3±7.9	29.9±0.5 ^a	27.8±0.5 ^b	28.9±0.4	55.2±0.4 ^a	50.3±0.4 ^b	52.7±0.4
L	78.4	55.4	47.0	1.12	1.55	0.94	0.73	0.81	0.33
C	11.85	10.02	11.18	2.25	1.40	2.81	0.79	0.97	0.87
GM		61.2			1.19			0.53	
GY		66.5			1.32			0.75	

M: Mean; L: LSD (5%); C: CV (%); GM: General Mean; GY: G × Year

Table 6. Sieve values of oat genotypes for >2.2 mm, 2.2-2.0 mm and 2.0-1.5 mm

G. No	2.2 mm Sieve Top (>2.2) (%)			Sieve Between 2.2-2.0 mm (2.2-2.0) (%)			2.0-1.5 mm The range (2.0-1.5) (%)		
	2019-20	2020-21	2019-21	2019-20	2020-21	2019-21	2019-20	2020-21	2019-21
1	57.1±0.6 ^c	68.4±1.2 ^d	62.8±2.6 ^e	24.2±0.3 ^e	17.2±1.1 ^{hi}	20.7±1.6 ^f	18.8±0.5 ^f	14.3±0.3 ^h	16.6±1.1 ^f
2	67.3±1.1 ^b	74.6±0.6 ^{bc}	71.0±1.7 ^b	20.1±0.4 ^{f-h}	16.4±0.7 ^{ij}	18.3±0.9 ^g	12.0±0.8 ^{hi}	8.7±0.4 ^j	10.4±0.8 ^h
3	48.3±2.1 ^d	60.9±1.0 ^g	54.6±3.0 ^g	19.0±0.8 ^h	13.8±0.7 ^{kl}	16.4±1.2 ^{hi}	31.4±1.7 ^c	25.1±0.4 ^f	28.3±1.6 ^c
4	39.0±1.8 ^e	42.2±0.8 ⁱ	40.6±1.1 ^j	32.2±1.3 ^{bc}	35.1±0.3 ^b	33.6±0.9 ^{ab}	28.2±2.6 ^d	22.5±0.6 ^g	25.3±1.7 ^d
5	22.2±1.5 ^f	15.5±0.5 ^l	18.8±1.7 ^l	27.2±1.1 ^d	38.4±0.6 ^a	32.8±2.6 ^b	49.0±2.0 ^a	45.4±0.6 ^c	47.2±1.2 ^b
6	21.4±0.4 ^f	26.3±0.4 ^k	23.9±1.1 ^k	29.9±0.3 ^c	18.4±0.4 ^h	24.1±2.6 ^{de}	45.8±1.2 ^b	51.3±0.6 ^b	48.6±1.4 ^b
7	66.2±1.6 ^b	64.8±1.6 ^{ef}	65.5±1.1 ^d	21.8±0.8 ^{e-g}	23.8±0.7 ^{ef}	22.8±0.7 ^e	11.7±0.7 ^{hi}	11.2±0.9 ^j	11.5±0.5 ^h
8	45.4±0.7 ^d	66.5±1.4 ^{de}	55.9±4.8 ^g	32.8±1.0 ^b	21.3±0.8 ^g	27.1±2.6 ^c	20.9±0.3 ^{ef}	12.0±0.7 ^j	16.4±2.0 ^f
9	57.5±1.5 ^c	62.6±1.3 ^g	60.1±1.4 ^f	22.4±0.8 ^{ef}	23.1±0.6 ^{fg}	22.8±0.5 ^e	20.0±0.7 ^{ef}	14.1±0.7 ^h	17.1±1.4 ^f
10	65.9±1.2 ^b	38.0±0.7 ^j	52.0±6.3 ^h	21.8±0.3 ^{ef}	28.6±0.3 ^d	25.2±1.5 ^d	12.0±0.9 ^{hi}	33.0±0.8 ^d	22.5±4.7 ^e
11	72.1±1.1 ^a	76.3±0.1 ^b	74.2±1.1 ^a	18.5±0.4 ^h	15.6±0.6 ^{ij}	17.1±0.7 ^{g-i}	9.4±0.6 ⁱ	7.9±0.6 ^j	8.7±0.5 ⁱ
12	64.2±1.7 ^b	72.5±0.4 ^c	68.4±2.0 ^c	20.4±0.6 ^{f-h}	15.1±0.2 ^{jk}	17.8±1.2 ^{gh}	14.4±1.0 ^{gh}	12.2±0.4 ⁱ	13.3±0.7 ^g
13	65.7±1.4 ^b	79.8±0.3 ^a	72.8±3.2 ^{ab}	19.2±0.4 ^{gh}	13.0±0.0 ^l	16.1±1.4 ⁱ	13.8±1.1 ^{gh}	7.1±0.3 ^{jk}	10.4±1.6 ^h
14	65.7±3.8 ^b	81.2±1.3 ^a	73.4±3.9 ^a	18.4±1.8 ^h	12.8±0.8 ^l	15.6±1.5 ⁱ	15.3±1.9 ^g	5.7±0.5 ^k	10.5±2.3 ^h
15	45.1±0.6 ^d	45.1±0.7 ^h	45.1±0.4 ⁱ	31.1±1.1 ^{bc}	25.0±0.3 ^e	28.1±1.5 ^c	22.6±1.2 ^e	29.4±0.5 ^e	26.0±1.6 ^d
16	17.2±1.1 ^g	63.0±0.5 ^m	11.8±2.5 ^m	37.6±0.6 ^a	31.2±0.5 ^c	34.4±1.5 ^a	43.0±1.0 ^b	60.9±0.3 ^a	52.0±4.0 ^a
M	51.3±2.7 ^b	55.1±3.7 ^a	53.2±2.1	24.8±1.0 ^a	21.8±1.1 ^b	23.3±0.7	23.0±1.8	22.6±2.7	22.8±1.5
L	4.10	2.70	2.40	2.54	1.81	1.53	2.93	1.65	1.65
C	4.80	2.94	3.92	6.14	4.98	5.67	7.64	4.38	6.26
GM		2.23			0.44			2.30	
GY		3.40			2.16			2.33	

M: Mean; L: LSD (5%); C: CV (%); GM: General Mean; GY: G.× Year

There statistically significant difference were determined among the 2.2-2.0 mm sieve values of the genotypes and years, genotypes and genotype x year interaction ($p<0.01$) (Table 4). In the study, the 2.2-2.0 mm sieve value of the genotypes varied between 18.4-37.6% (24.8) in the 1st year, 12.8-38.4% (21.8) in the 2nd year, and 15.6-34.4% (23.3) compared to the two-year average. According to the two-year average, genotype number 16 reached the highest 2.2-2.0 mm sieve value with 34.4%, and genotype number 14 reached the lowest 2.2-2.0 mm sieve value with 15.6% (Table 6). In the first year, the 2.2-2.0 mm sieve values of the genotypes were higher than the second year, parallel to the grain yield, 1000 grain weight and hectolitre weights. Brunova et al. (2015) reported that the 2.2-2.5 mm sieve value varied between 30.96% and 34.85% and the <2.2 mm sieve value varied between 5.02% and 10.36% in 19 oat varieties as a result of a 3-year study.

There were found that statistically significant differences among the genotypes, 2.0-1.5 mm sieve values and genotype x year interaction ($p<0.01$), but it was not statistically significant between years. While a positive and significant relationship was determined between 1000 grain weight and >2.2 mm sieve value ($r=0.5710$) of the genotypes, a negative and significant relationship was determined between >2.2 mm sieve value and 2.2-2.0 mm ($r=-0.7920^{**}$) and 2.0-1.5 mm sieve value ($r=-0.9613^{**}$). The 2.2 mm sieve value was also high in genotypes with high thousand grain weight. Although the 1000 grain weight of genotype no. 16 was high [29.2 - 34.0 (31.6) g] the 2.2 mm sieve value of some genotypes was low. The reason for this is that the grain of the variety is thin and long. While the 2.2 mm sieve value of the genotypes increases, the 2.2-2.0 mm and 2.0-1.5 mm sieve values decrease. In the study, the 2.0-1.5 mm sieve value of the genotypes changed between 9.4-49.0% (23.0) in the 1st year, 5.7-60.9% (22.6) in the 2nd year, and 8.7-52.0% (22.8) according to the two-year average. According to the

two-year average, genotype number 16 reached the highest 2.0-1.5 mm sieve value with 52.0%, and genotype number 11 reached the lowest 2.0-1.5 mm sieve value with 8.7% (Table 6).

Significant differences were found among genotypes, years, and genotype × year interactions ($p<0.01$) (Table 4). In the study, the oil ratio of the genotypes varied between 0.32-6.12% (3.49) in the 1st year, 0.16-6.64% (2.83) in the 2nd year, and 0.28-6.38% (3.20) compared to the 2 years average. According to the 2 years average, genotype 5 reached the highest oil ratio with 6.38%, and genotype 13 reached the lowest oil ratio with 0.28% (Table 7). The oil ratio of the genotypes in the first year was higher than in the second year. Since the high amount of oil, a significant portion of the fatty acids consist of unsaturated fatty acids and have high lipase activity, shortens the shelf life of oats and oat products, it is desired that the oil content is low in human nutrition. Since fat is a good source of energy, it is desired to have a high oil content in animal feeding. In previous studies, Şahin et al. (2019) observed that the oil ratio in oat genotypes varied between 3.23% and 8.09%, Topkara (2019) observed that the oil ratio of the genotypes varied between 6.76-8.24%, and Çiçek (2019) observed that the oil ratio varied between 5.20-6.51%. Howarth et al. (2021) reported that the average oil ratio in four winter oat varieties in different locations over four years varied between 6.48% and 7.83%, and that environmental conditions and diversity affected the oil ratio.

When the protein ratio was examined, statistically significant differences were observed among genotypes, years, and genotype × year interactions ($p < 0.01$) (Table 4). In the study, the protein ratio of the genotypes varied between 8.8-12.3% (10.2) in the first year, 10.1-13.0% (11.8) in the second year, and 9.4-11.8% (11.0) according to the two-year average. According to the two-year average, genotypes 13 and 14 reached the highest protein rate with 11.8%, and genotype 16 reached the lowest protein rate with 9.4% (Table 7).

Table 7. Oil and protein rates values of oat genotypes

G. No	Oil ratio (%)			Protein ratio (%)		
	2019-20	2020-21	2019-21	2019-20	2020-21	2019-21
1	5.9±0.1 ^a	4.2±0.2 ^c	5.1±0.4 ^b	10.1±0.1 ^{ef}	11.1±0.1 ^e	10.6±0.2 ^g
2	0.8±0.1 ^g	0.6±0.2 ^h	0.7±0.1 ^k	10.7±0.0 ^c	12.4±0.0 ^b	11.6±0.4 ^{bc}
3	5.7±0.1 ^a	3.2±0.2 ^d	4.5±0.6 ^{cd}	9.7±0.1 ^g	11.4±0.1 ^d	10.6±0.4 ^g
4	5.0±0.1 ^b	2.4±0.1 ^{ef}	3.7±0.6 ^{fg}	9.7±0.2 ^g	12.0±0.0 ^c	10.8±0.5 ^f
5	6.1±0.1 ^a	6.6±0.1 ^a	6.4±0.1 ^a	9.4±0.1 ^g	10.7±0.0 ^f	10.1±0.3 ^h
6	4.2±0.2 ^c	5.2±0.1 ^b	4.7±0.2 ^c	11.3±0.1 ^b	11.2±0.0 ^e	11.2±0.0 ^e
7	4.5±0.1 ^c	3.3±0.1 ^d	3.9±0.3 ^{ef}	9.5±0.2 ^g	11.3±0.1 ^{de}	10.4±0.4 ^g
8	0.9±0.2 ^g	3.3±0.1 ^d	2.1±0.5 ^{ij}	12.3±0.0 ^a	11.1±0.1 ^e	11.7±0.3 ^{ab}
9	3.8±0.2 ^d	4.2±0.1 ^c	4.0±0.1 ^{ef}	10.0±0.1 ^f	12.3±0.0 ^b	11.2±0.5 ^e
10	4.4±0.2 ^c	2.7±0.0 ^e	3.5±0.4 ^g	9.5±0.1 ^g	12.5±0.0 ^b	11.0±0.7 ^f
11	3.2±0.1 ^e	1.2±0.2 ^g	2.2±0.5 ⁱ	10.2±0.1 ^{ef}	12.3±0.2 ^b	11.3±0.5 ^{de}
12	3.5±0.2 ^{de}	2.5±0.2 ^{ef}	3.0±0.3 ^h	10.4±0.1 ^{de}	12.5±0.1 ^b	11.4±0.5 ^{cd}
13	0.3±0.0 ^h	0.2±0.1 ^{hi}	0.3±0.0 ⁱ	10.6±0.1 ^{cd}	13.0±0.1 ^a	11.8±0.5 ^a
14	0.8±0.1 ^g	0.2±0.0 ⁱ	0.5±0.1 ^{kl}	10.6±0.1 ^{cd}	12.9±0.1 ^a	11.8±0.5 ^a
15	1.5±0.1 ^f	2.2±0.3 ^f	1.9±0.2 ^j	10.9±0.1 ^c	12.6±0.1 ^b	11.7±0.4 ^{ab}
16	5.0±0.0 ^b	3.3±0.1 ^d	4.2±0.4 ^{de}	8.8±0.1 ^h	10.1±0.1 ^h	9.4±0.3 ⁱ
M	3.5±0.3 ^a	2.8±0.2 ^b	3.2±0.2	10.2±0.1 ^b	11.8±0.1 ^a	11.0±0.1
L	0.42	0.41	0.29	0.29	0.24	0.18
C	7.19	8.75	7.89	1.71	1.19	1.44
GM		0.29			0.10	
GY		0.41			0.26	

M: Mean; L: LSD (5%); C: CV (%); GM: General Mean; GY: G.× Year; *Means marked with the same letter are no different from each other. Ö: significant

In the first year, the protein ratio of the genotypes was lower than in the second year. It is desired to have a high protein ratio in both human nutrition and animal nutrition. In previous studies, Khan et al. (2014) found that the protein ratio in oat genotypes varied between 9.72% and 13.84% in Pakistani conditions. Kahraman et al. (2017a) reported that the protein ratio in oat genotypes tested in different locations for 2 years varied between 9.0% and 15.2%, emphasizing the value of protein in human and animal nutrition. Halil and Uzun (2019) found that the protein ratio of oat genotype varied between 5.67% and 8.44% in Bursa ecological conditions. Howarth et al. (2021) reported that the protein ratio in four winter oat varieties in different locations for four years was affected by the variety and environmental conditions, and reported that the average protein ratio varied between 7.77% and 12.33%, which is consistent with our findings. Güngör et al. (2023) stated that the protein ratio in nine oat genotypes at different locations over two-year study ranged from %12.56 to 15.10.

According to the results of the correlation analysis among the examined traits, a positive and significant relationship were determined between 1000 grain weight in both hectoliter weight ($r = 0.6473^{**}$) and >2.2 mm sieve value ($r = 0.5710$). A negative and significant relationship was determined between >2.2 mm sieve value in both 2.2-2.0 mm ($r = -0.7920^{**}$) and 2.0-1.5 mm ($r = -0.9613^{**}$). In previous studies, Sarı (2012) found a positive and significant relationship between hectoliter weight and 1000 grain weight, oil content and protein content among the traits examined in oat genotypes. Naneli and Sakin (2022) stated that there was an insignificant relationship between protein amount, 1000 grain weight and hectoliter weight among the traits examined in oat genotypes.

Conclusion

According to the results of 2 years, the effect of growing seasons on the sieve value for 2.0-1.5 mm was not statistically significant. All of the other traits were statistically affected by years, genotypes and genotype x year interaction. According to means of 2 years, the highest grain yield values were achieved by the Kahraman, Elmas and genotype 7. When the genotypes were evaluated in terms of yield and quality characteristics according to the obtained data, Kahraman and Elmas cultivars were found to be suitable for oat cultivation in Kahramanmaraş ecological conditions. And also, 7, 8 and 11 advanced lines have been deemed appropriate for registration based on to all traits investigated in the same region.

Declarations

Ethical Approval Certificate

Our study does not require Ethics Committee approval.

Author Contribution Statement

Authors declares the contribution of the authors is equal.

Conflict of Interest

The authors declare no conflict of interest.

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